

DIGITAL TWIN-BASED PREDICTIVE MAINTENANCE FOR WATER PUMPS IN SMART IRRIGATION SYSTEMS

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Abstract—Water pumps are among the most critical and failure-prone components in smart irrigation systems, directly influencing water efficiency and crop productivity. This study proposes an end-to-end Digital Twin-based predictive maintenance framework specifically targeting water pumps in smart irrigation environments, integrating real-time sensing, cloud-based monitoring, anomaly detection, and Remaining Useful Life (RUL) prediction. An ESP32-based hardware prototype was developed to acquire pump-related sensor data and transmit it to a cloud platform via WiFi for continuous condition monitoring. The Digital Twin serves as a virtual replica of the physical pump, continuously updated using sensor streams to reflect operational states, monitor degradation patterns, and support maintenance planning before actual failure occurs. The dataset combines publicly available real-world sensor data with simulated fault-condition samples to compensate for limited availability of real failure events. Feature selection reduced dimensionality from 52 to 9 significant sensors, improving model efficiency and interpretability. A Logistic Regression model with threshold optimisation and class-balanced learning achieved a weighted F1-score of 0.95 and a test accuracy of 93.7% for anomaly detection. For RUL prediction, a Random Forest Regressor achieved a test R^2 of 0.9354 and a Mean Absolute Error of 33.12 hours. The novelty of this work lies in the integration of Digital twin technology, machine learning and an end-to-end hardware-to-cloud architecture within an irrigation-specific predictive maintenance application. The proposed framework provides a scalable and practical solution for reducing pump downtime and supporting sustainable smart agriculture.

Index Terms—Digital Twin, Predictive Maintenance, Smart Irrigation, Water Pump, Anomaly Detection, Remaining Useful Life, IoT